

Did GRPO recover what the CoT stage cost us?

Qwen3.5-VL-0.8B, evaluated at each stage of the pipeline — SFT-only → +CoT → +GRPO — on visual grounding (VG_DOIR_RSVG_test) and five remote-sensing classification benchmarks. The CoT stage traded accuracy for reasoning; GRPO's job was to win it back, ideally past the SFT-only ceiling. Greedy decoding (temp=0.0), recot-eval / vLLM.

Visual grounding — VG_DOIR_RSVG_test

7,500 boxes · IoU@0.5. GRPO recovered a little of what CoT lost, but landed short of the SFT-only ceiling on every size bucket except large objects.

Stage	Precision	Small	Medium	Large
SFT-only	0.7565	0.4633	0.7175	0.8736
+ CoT	0.7223	0.4217	0.6811	0.8429
+ GRPO	0.7259 (− 3.06pp)	0.4180 (− 4.53pp)	0.6878 (− 2.97pp)	0.8473 (− 2.63pp)

Classification — 5 benchmarks

Accuracy. A more mixed picture: GRPO overshot the SFT-only baseline on three benchmarks — including a clean +2.32pp swing on NWPU-RESISC45 — but landed slightly below even the CoT stage on AID, its one regression.

Benchmark (n)	SFT-only	+ CoT	+ GRPO	vs SFT-only	Verdict
AID (3,000)	0.9157	0.8977	0.8940	− 2.17pp	regressed
NWPU-RESISC45 (3,150)	0.9057	0.8898	0.9130	+ 0.73pp	beat SFT
WHU-RS19 (1,000)	0.9630	0.9550	0.9660	+ 0.30pp	beat SFT
METER-ML (565)	0.6903	0.6389	0.6513	− 3.90pp	partial recovery
SIRI-WHU (2,400)	0.7296	0.7308	0.7354	+ 0.58pp	beat SFT

SFT-only = qwen35-{vg,cls}-merged · +CoT = qwen35-cot-{vg,cls}-merged · +GRPO = this run's output (Qwen-VL-Series-Finetune/output/grpo/). Deltas throughout are GRPO vs SFT-only — the recovery target.